

When is specialization worth its cache footprint?

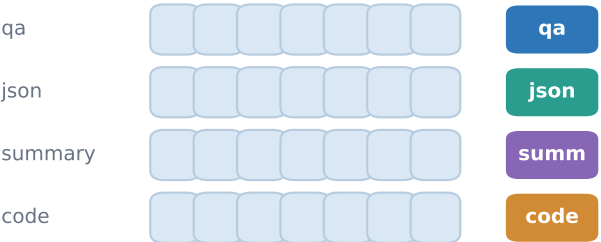
Specialist adapters improve task quality; shared-prefix reuse decides whether that quality is cheap enough to serve.

1. The hidden cost

Routing to a specialist can improve quality, but it also changes the KV-cache namespace.

standard LoRA

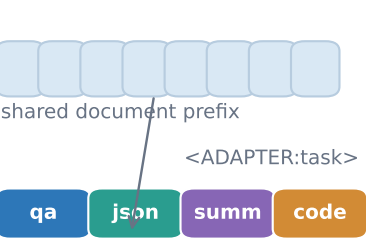
same prefix cached once per adapter



prefix footprint scales with adapter count

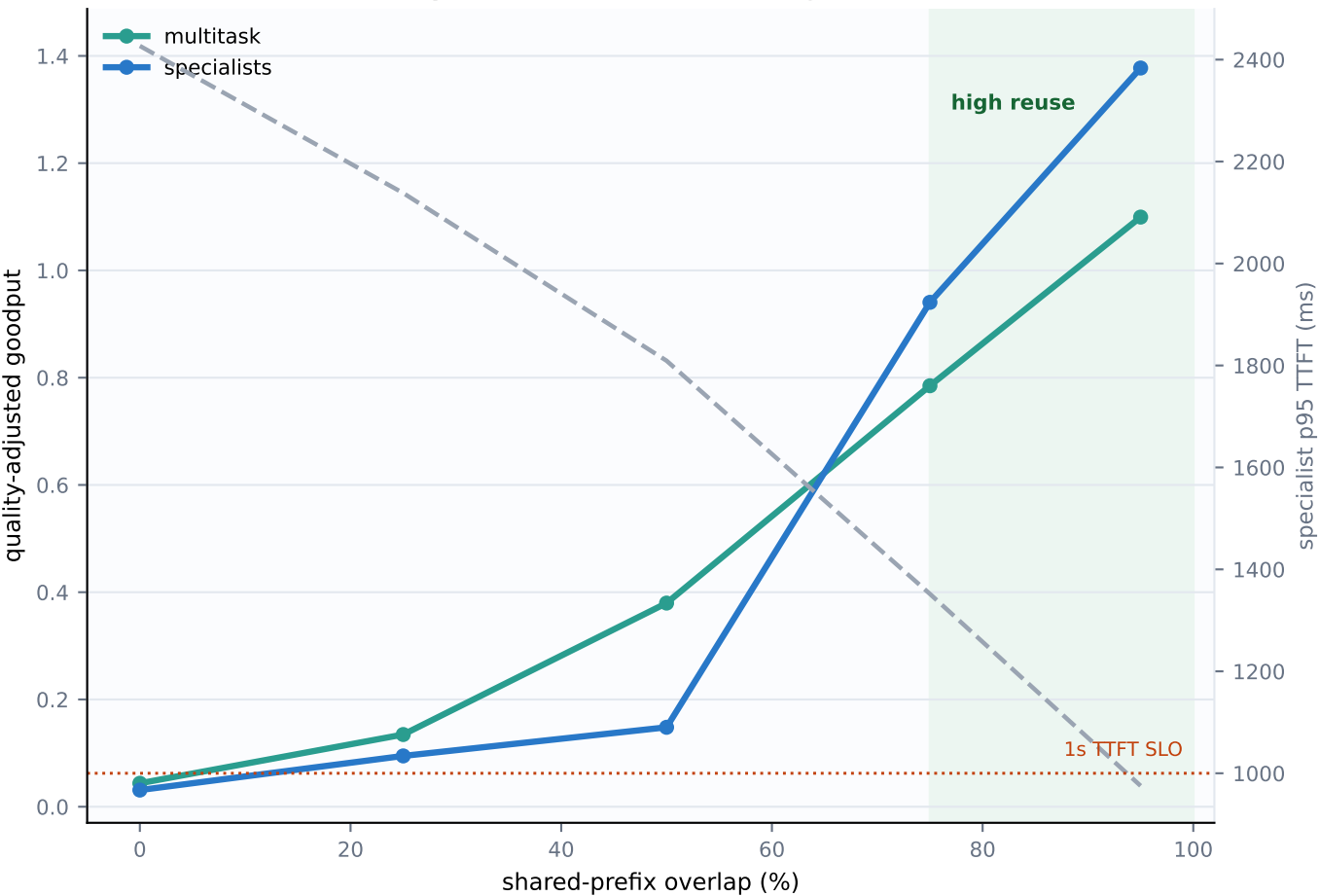
late specialization

shared prefix, adapter-specific tail



prefix footprint stays nearly flat

2. Specialization pays when overlap is high



Decision rule: specialize only when quality gain exceeds the cache-footprint and SLO cost.